

Advancing Shock Simulation: Applying the Smallwood Algorithm to Explicit Finite Element Models

Alexis Castel, Ph.D. – Keysight Technologies

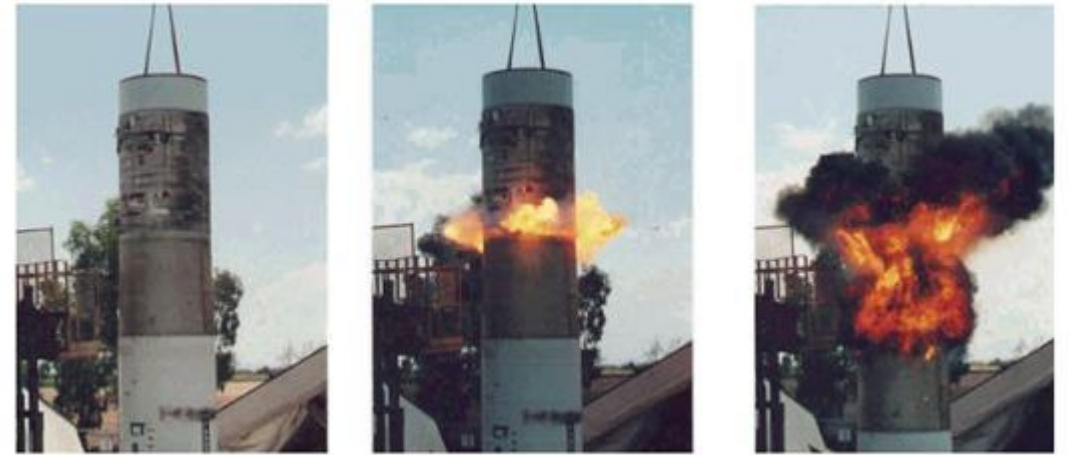
Agenda

Subtitle

- Background & Motivation
- Shock Response Spectrum (SRS) essentials
- Case study using ShockSat dataset
- Smallwood SRS synthesis algorithm
- Explicit finite element approach
- Results & validation
- Key takeaways

Background: Shock Environments in Engineering

- Pyro-shocks in spacecraft stage separation
- Drop & impact events
- Still a source of test or launch failure
- Need for reliable prediction early in design



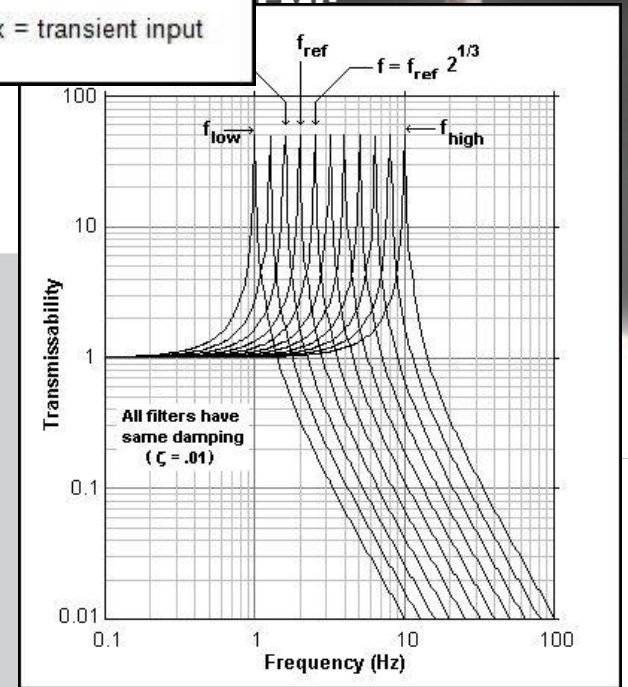
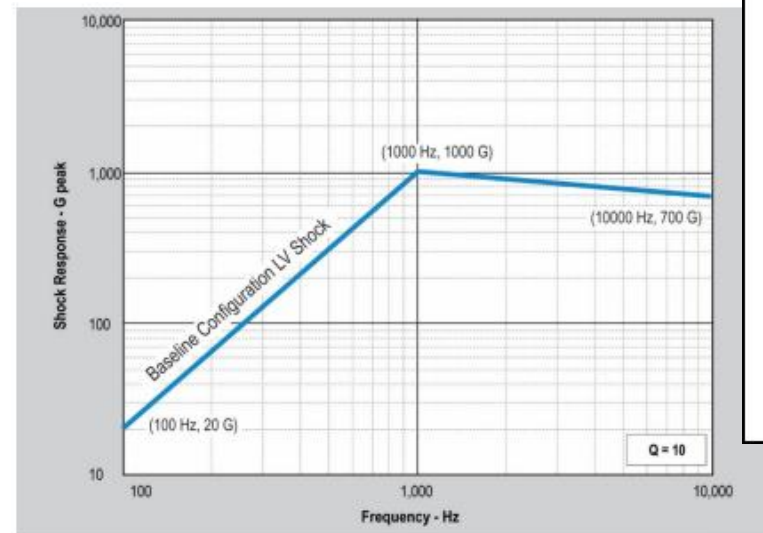
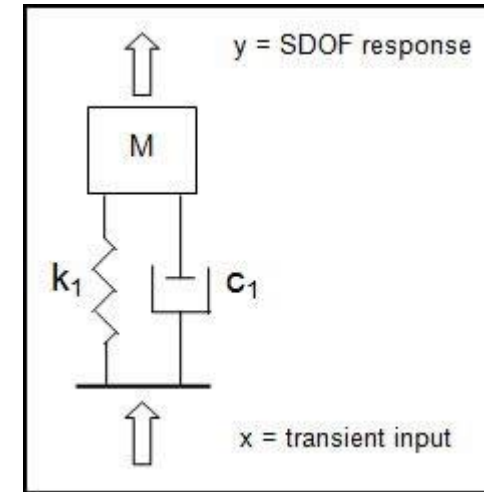
Source: nasa.gov

PYROTECHNIC SHOCK FLIGHT FAILURES*

By C. J. Moening, The Aerospace Corporation

Shock Response Spectrum (SRS)

- Peak acceleration vs natural frequency graph
- Derived from single-DOF response
- Standard test & design metric

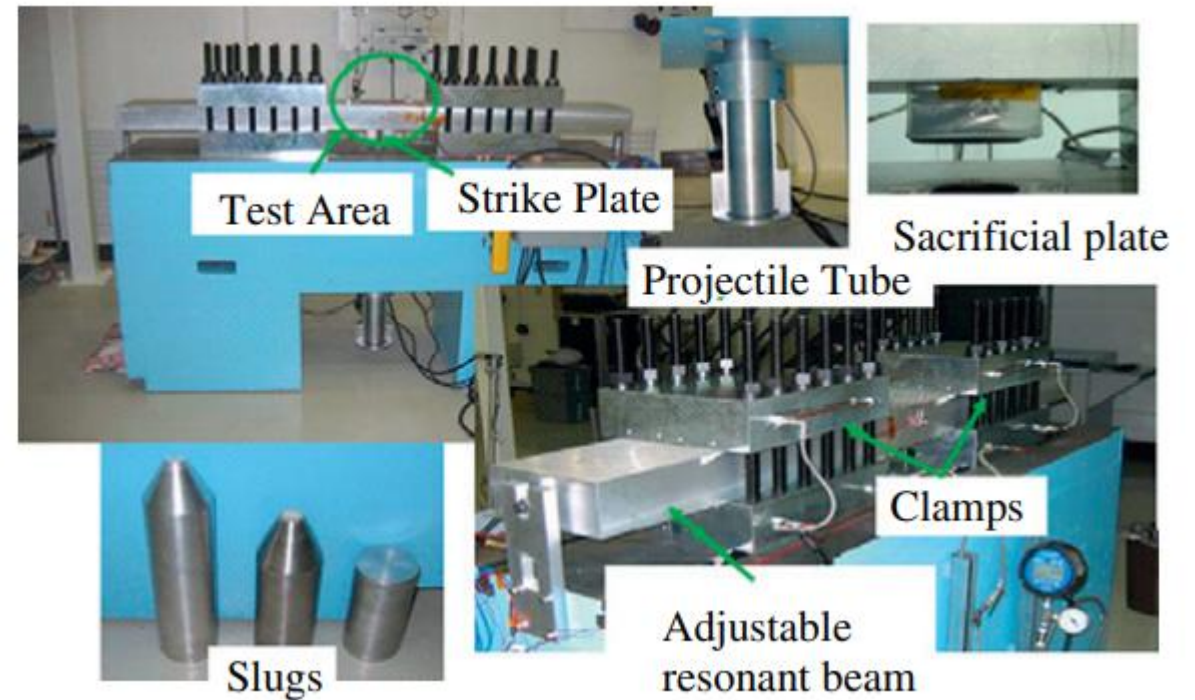


Source: dataphysics.com

*5% damping assumed for all SRS in this presentation

Physical Ways to Realize an SRS

- Hammer swings on beams
- Air-gun lug shots
- Pyro-shock simulators
- Electrodynamic shaker 'shock' tests



Source: Lee et al., Review of pyroshock wave measurement and simulation for space systems, 2012

Prediction Of Pyroshock Environments

Commonly accepted methods

NASA-STD-7003A

APPENDIX A

PREDICTION OF PYROSHOCK ENVIRONMENTS

There are three general ways to predict or estimate the response at various locations on a structure induced by a pyrotechnic device as follows:

- a. Analytical models.
- b. Direct measurements.
- c. Extrapolations from previous measurements.



All three methods use an SRS to define the input shock load.

The Challenge

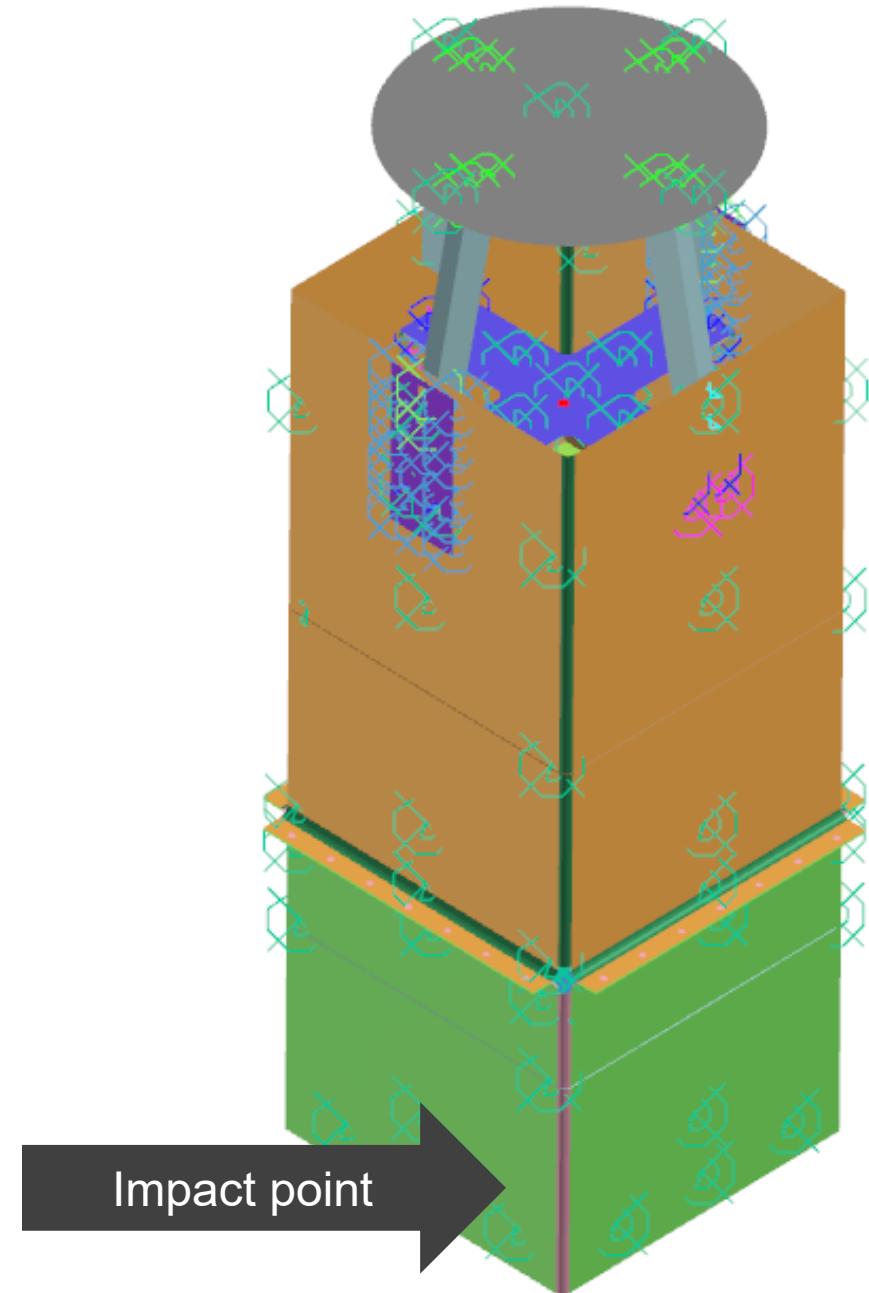
- Underdetermined inverse problem
- Many waveforms share the same SRS
- Time-domain history needed for simulation & control



Objective: Using SRS to define input excitation for numerical simulation models.

ShockSat Dataset

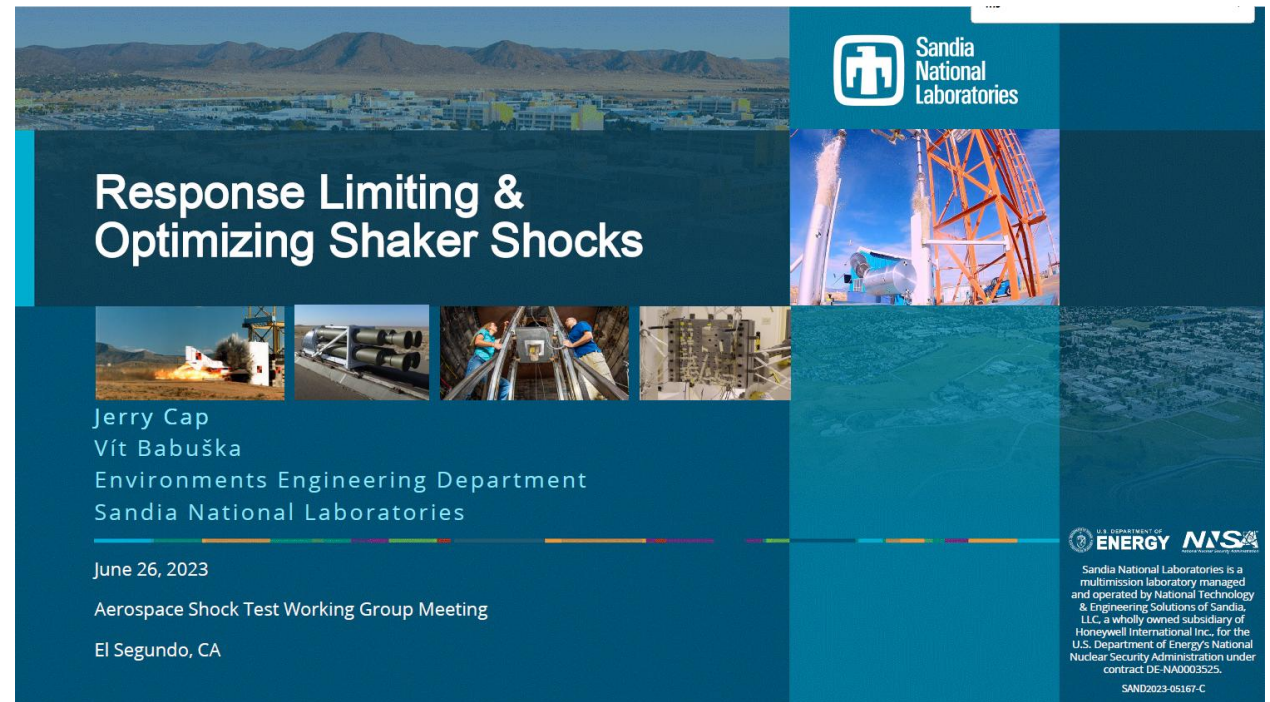
- NASA public benchmark (2021-2022) for shock prediction
- Satellite-like structure with bolted joints
- 108 accelerometer channels, hammer & pyro tests
 - Time domain acceleration response
 - Time domain impact force
- CAD and Correlated FEM openly available
 - Allows for numerical simulation



Smallwood Algorithm – Overview

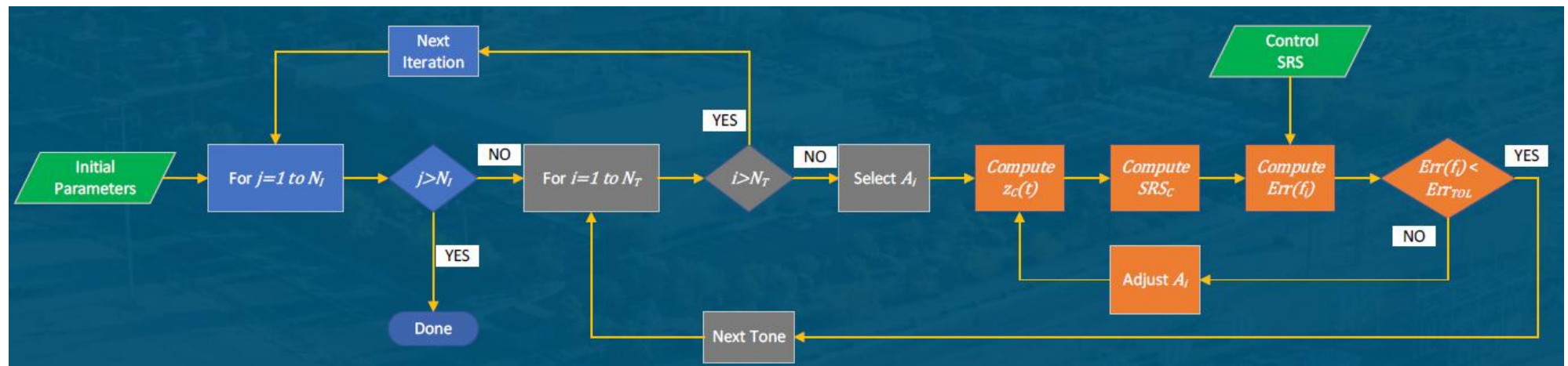
- Developed at Sandia National Labs (1970-1980s)
- Synthesizes time-history matching target SRS
- Uses sum of damped sinusoids

2023 Shock working group
presentation



Mathematical Formulation

- $z(t) = \sum_{i=1}^N A_i e^{-2\pi f_i \zeta_i t} \sin(2\pi f_i t)$
- Amplitudes iteratively adjusted to hit SRS
- Constraints for shaker displacement & velocity

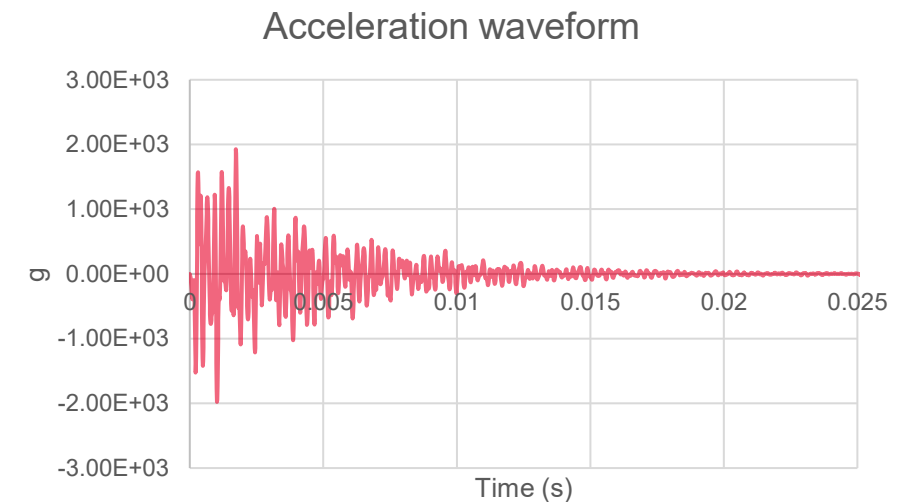
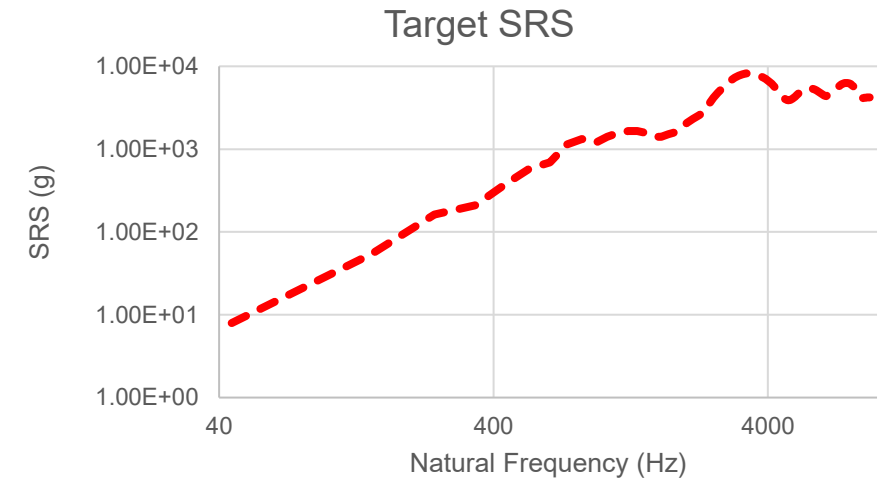


Source: response limiting and optimizing shaker shock (Sandia)

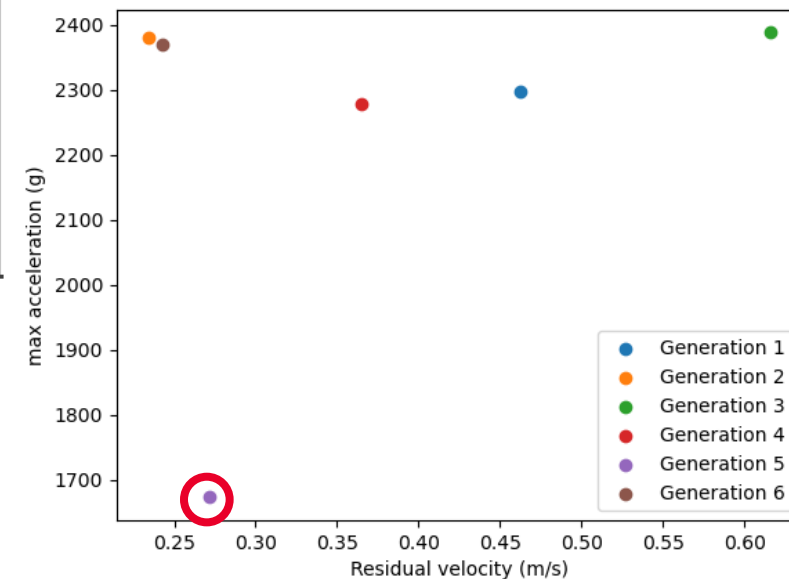
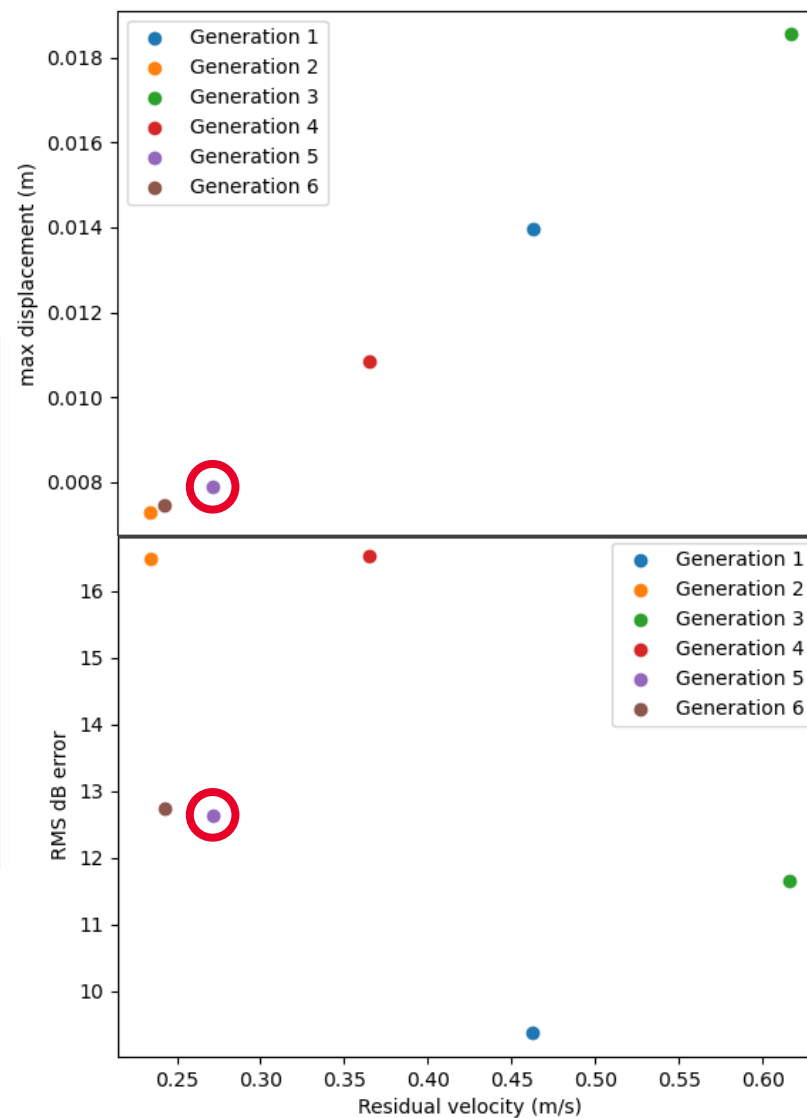
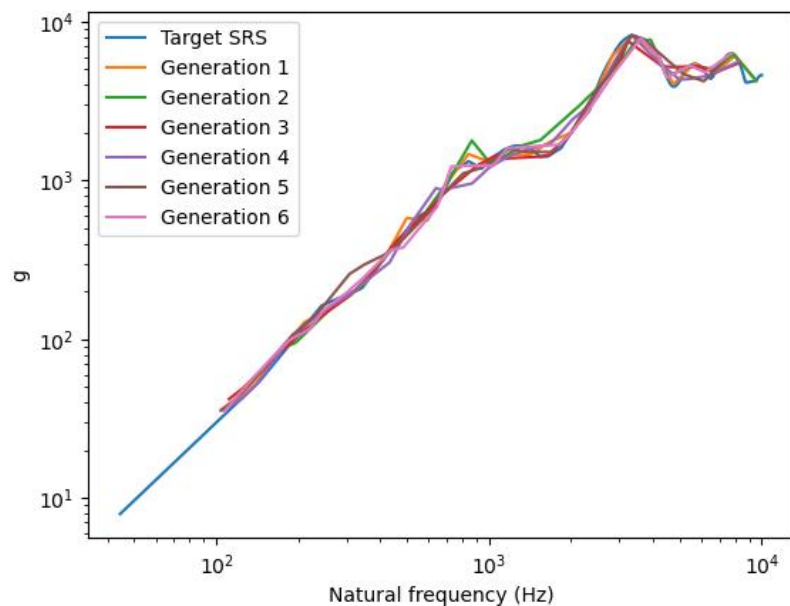
Smallwood Implementation Flow

Objective: SRS → Realistic time-domain base acceleration waveform

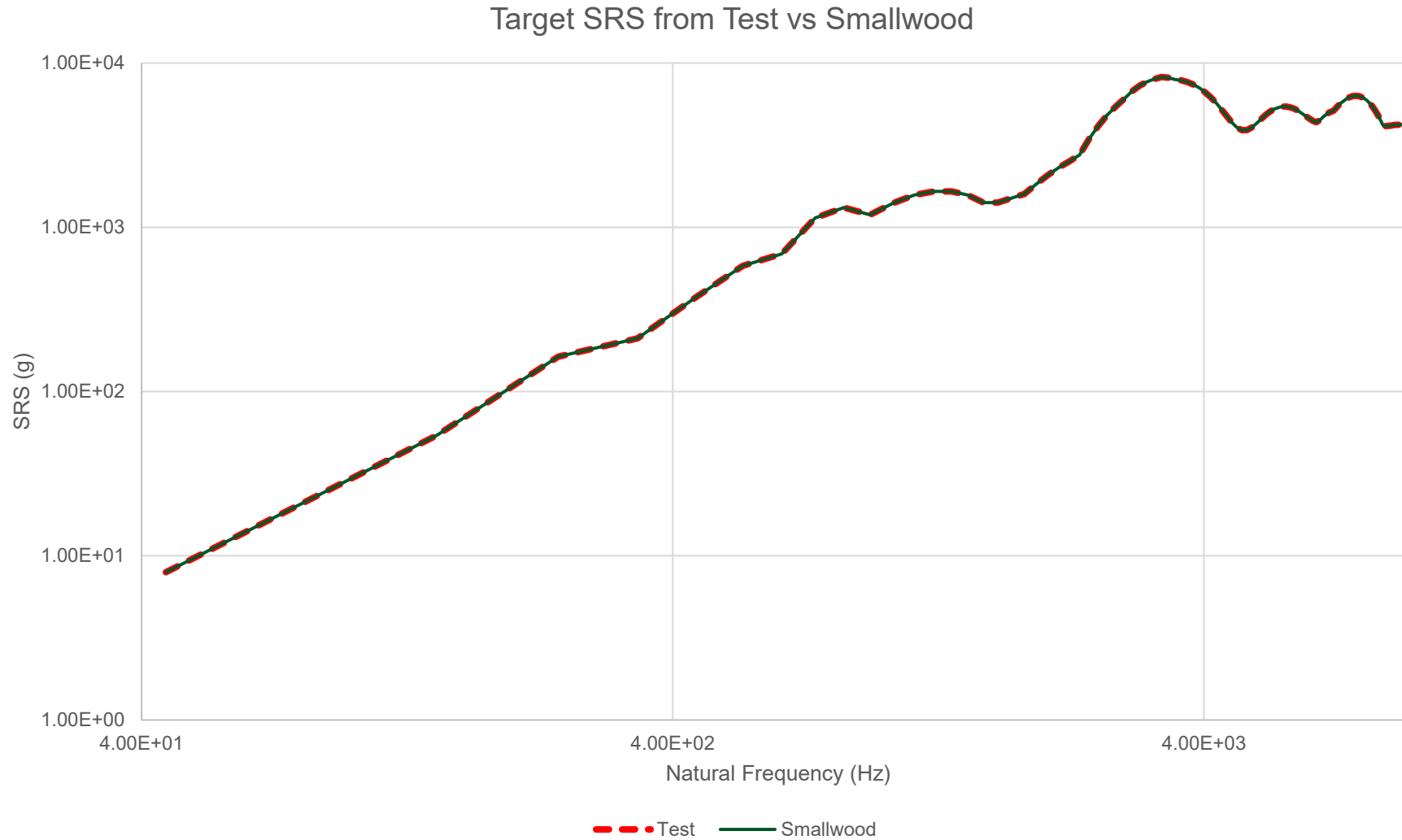
- Input: target SRS & damping
- Select frequency set
 - Influenced by duration of shock signal
- Compute modal responses
- Iteratively adjust amplitudes to match SRS.
Optimize for:
 - Maximum acceleration
 - Maximum displacement
 - Residual velocity
- Export acceleration waveform



Smallwood Implementation



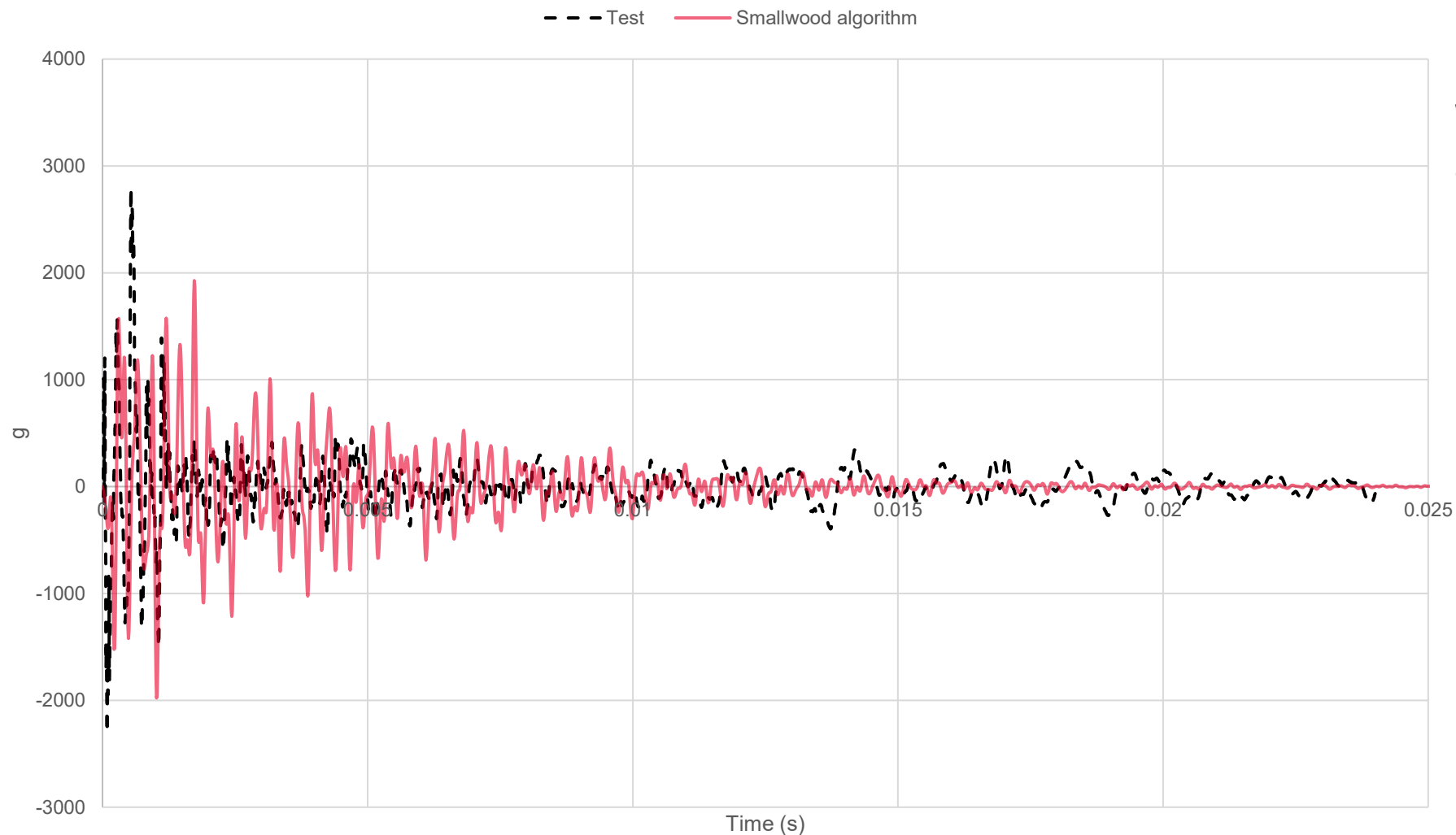
Target SRS from Test vs Smallwood



Algorithm has no issue matching a complex shape SRS:

- No physical limitation of a shaker
- Need to watch out for the waveform to be realizable

Test waveform vs Smallwood



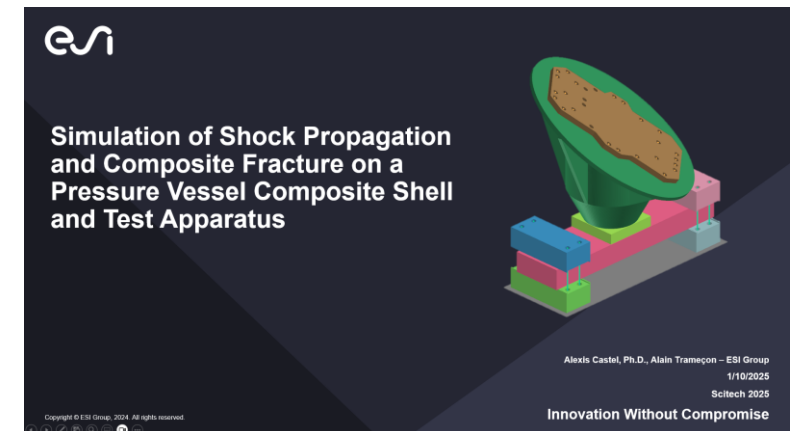
Waveform is different but shows a similar trend and amplitude.

Explicit Finite Element Modeling

- Central-difference explicit time integration → smaller time step
- Handles highly transient, nonlinear response
- Published examples:



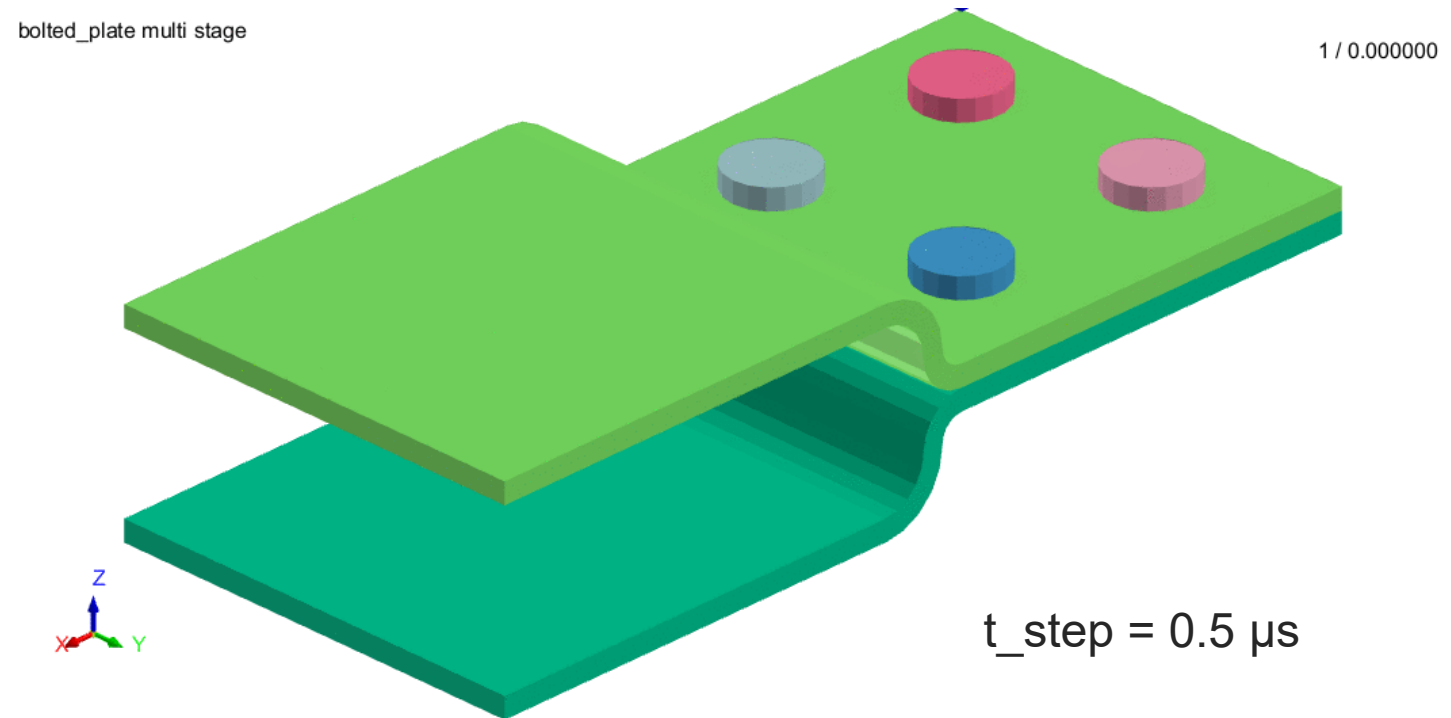
SCLV 2024



SciTech 2025

Why Explicit for Shock?

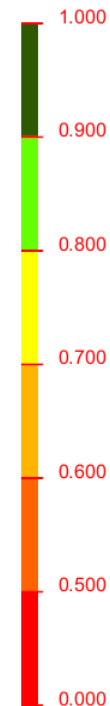
- Time steps $\ll$ shock rise time
- Captures contact & plasticity
- No modal reduction required



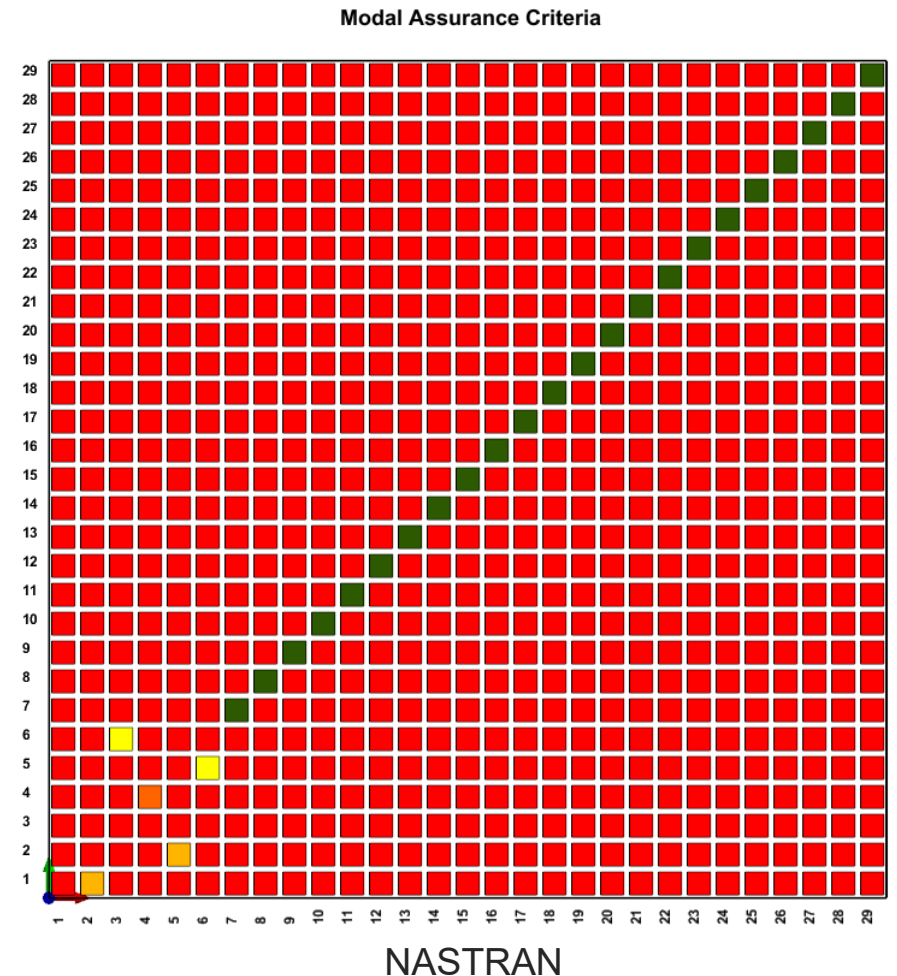
ShockSat in Virtual Performance Solution (VPS)

- Model translation
- MAC is healthy
- Losses modeled with a combination of:
 - Friction at contacts
 - Rayleigh damping

MAC VALUE



VPS

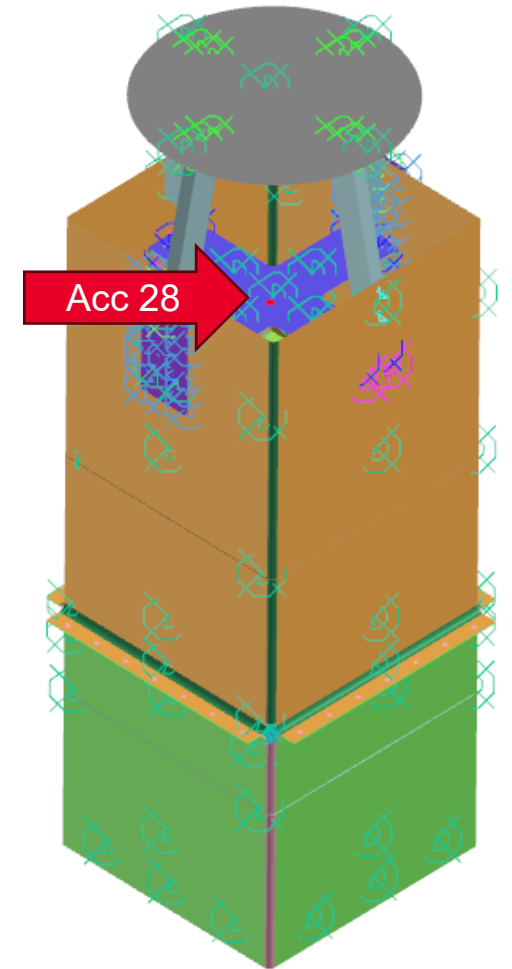
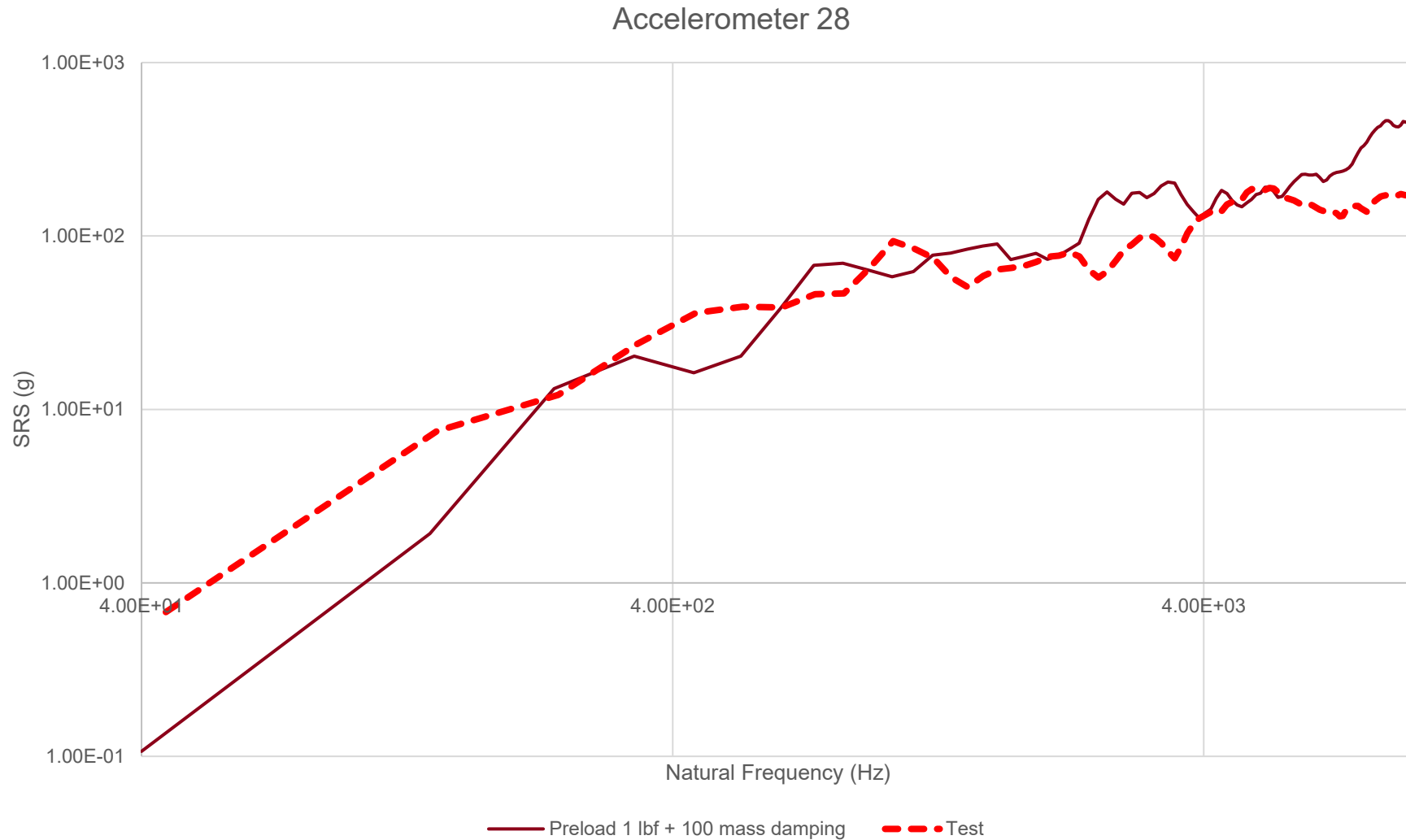


Putting it all together

- Target SRS → Smallwood → accel vs time
- Apply as base excitation or nodal load to FE model
- Solve explicit simulation
- Post-process SRS at response points

VPS response

Using measured hammer force

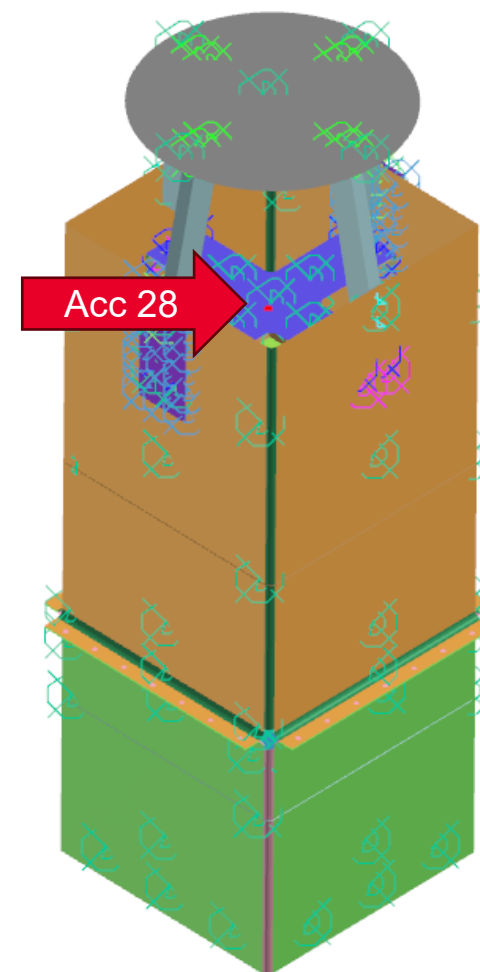
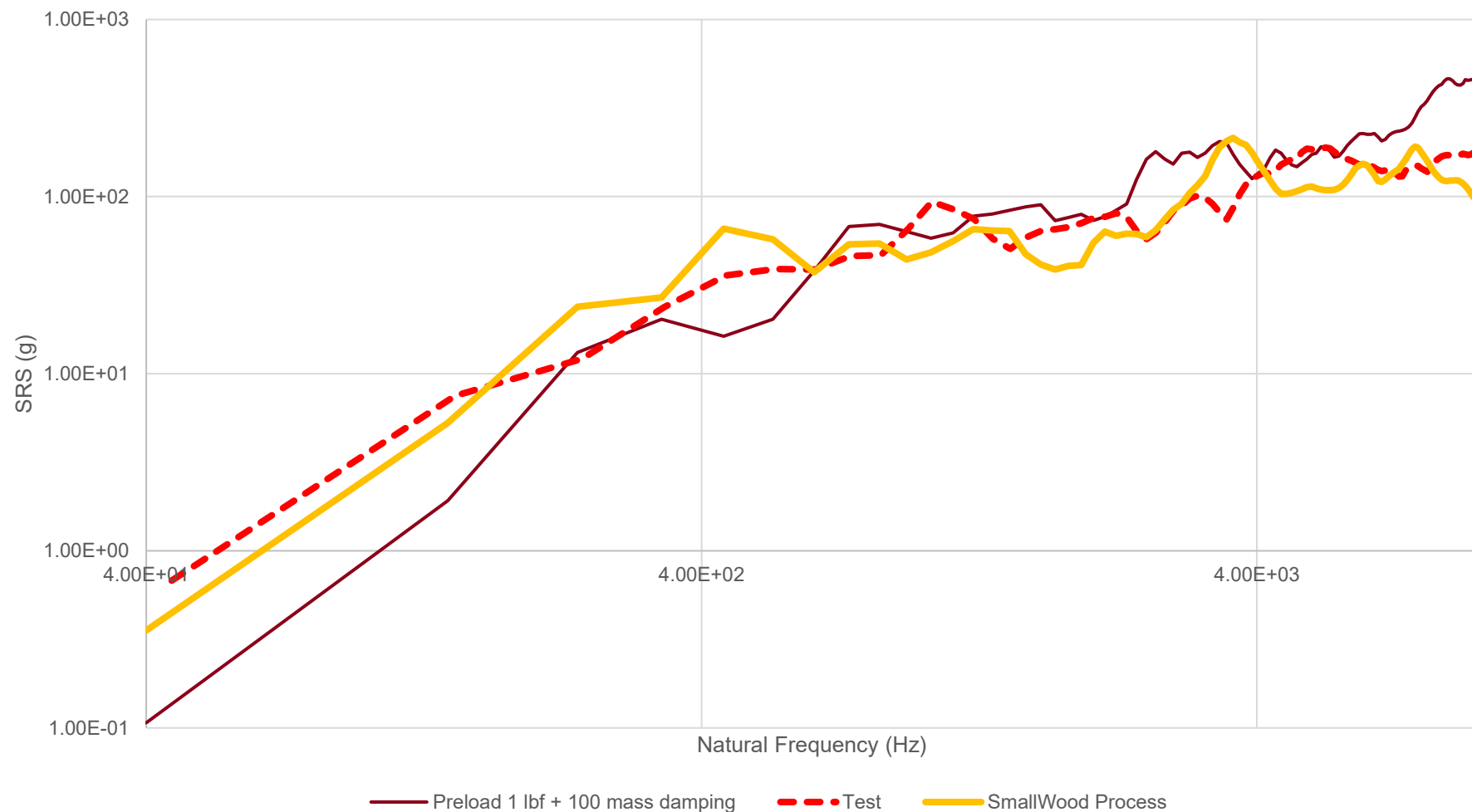


(Time step = 0.5 μ s)

VPS response

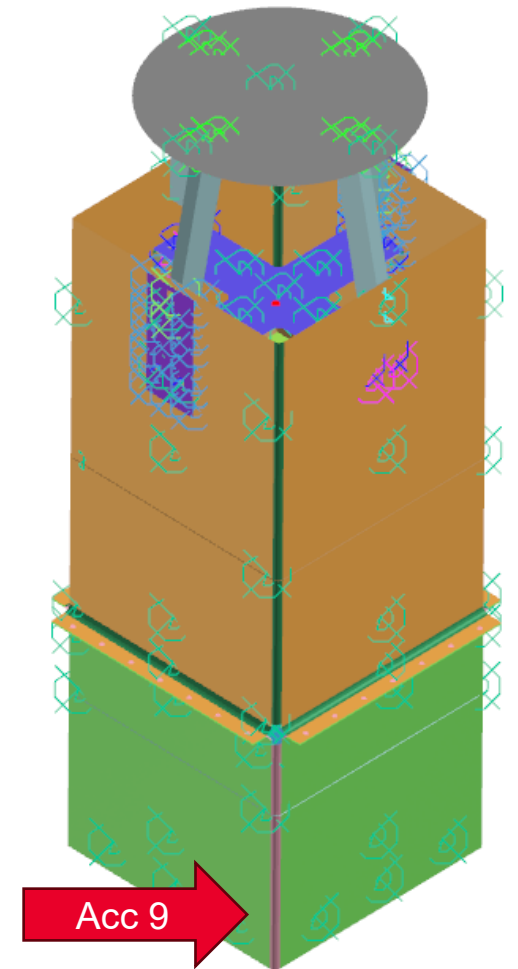
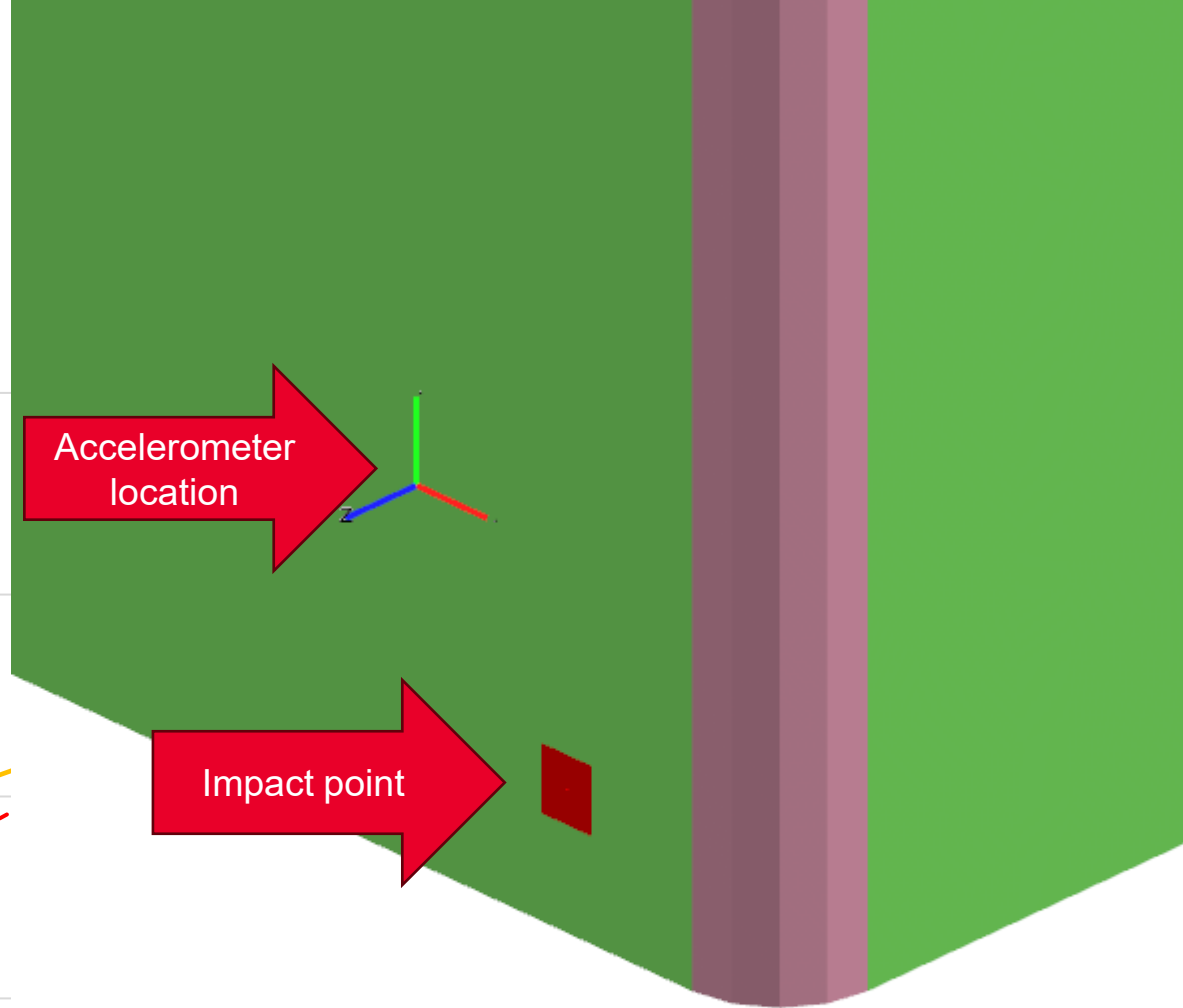
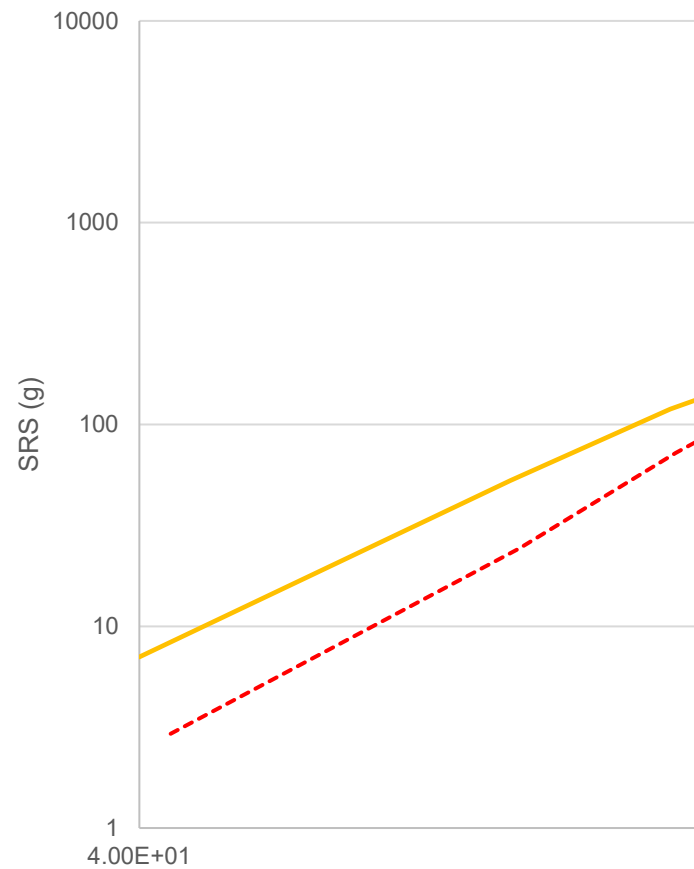
Using imposed acceleration at impact point from Smallwood algorithm

Accelerometer 28

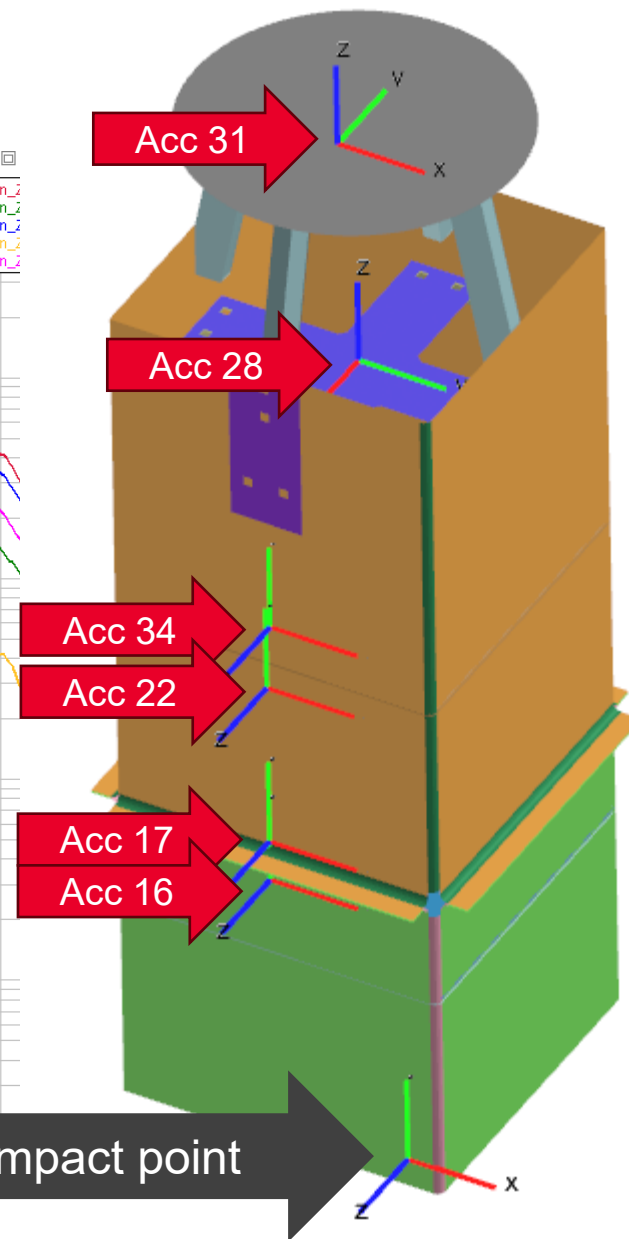
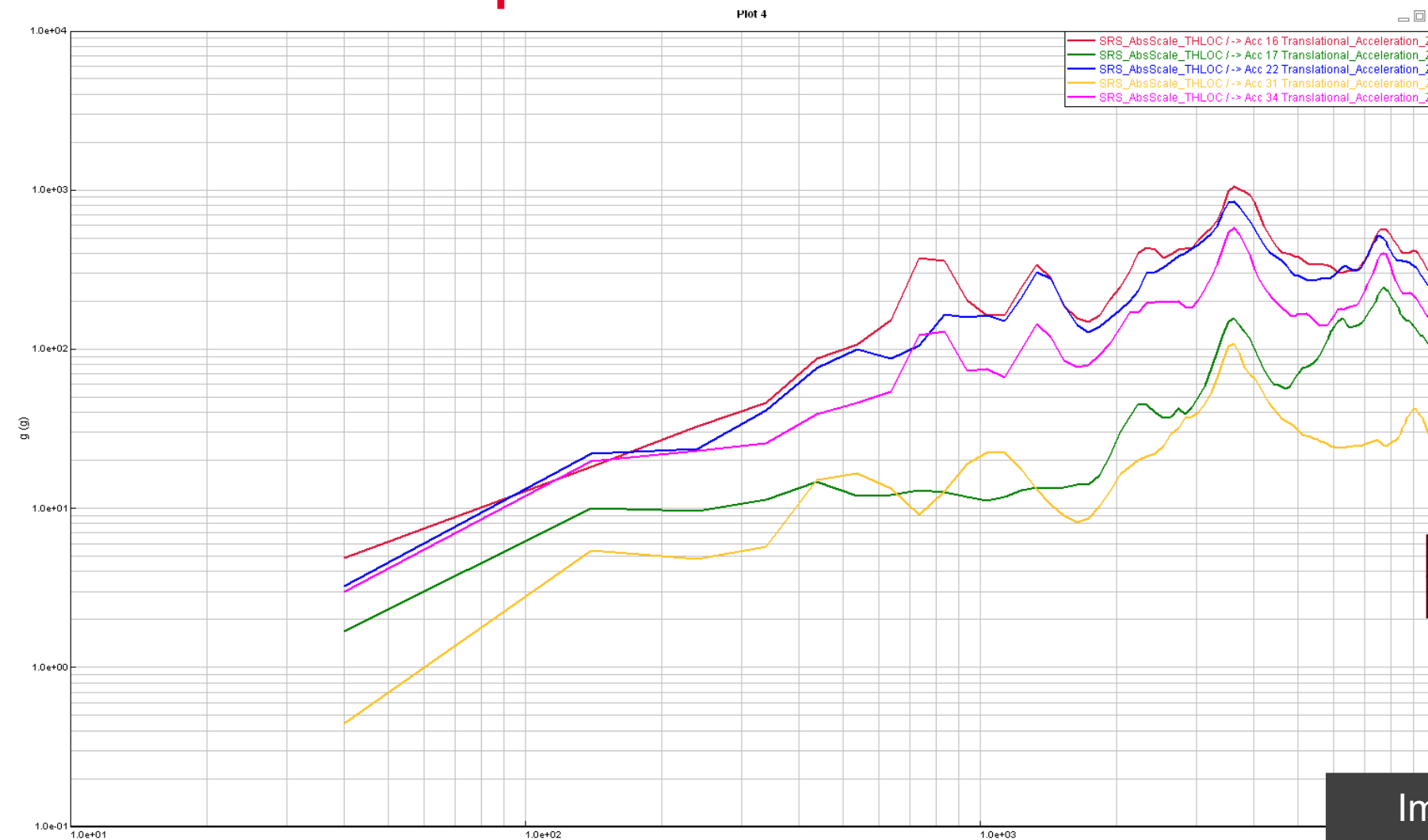


VPS response

Impact point differences



Additional data points



Limitations & Considerations

Smallwood:

- Non-uniqueness of synthesized signal
- SRS ignores phase & duration details

Explicit FEM:

- Accurate damping/contact assumptions required
- Large explicit models have a computational cost

Conclusions & Q&A

- Smallwood + explicit FE = practical shock prediction
- Public ShockSat case validates approach
- Leverages existing explicit FE models
- Supports digital qualification early
- Compatible with shaker hardware limits
- Ready for industrial integration

Thank you

